CSc. 451-2075✡

Tribhuvan University

Institute of Science and Technology

2075

✡

Bachelor Level / Forth Year /Eighth Semester/Science

Computer Science and Information Technology-(CSc. 451)

(Data warehousing and Data mining)

Full Marks: 60
Pass Marks: 24
Time: 3 hours.

Candidates are required to give their answers in their own words as for as practicable.
The figures in the margin indicate full marks.

Group A

Attempt any two questions. [2×10=20]

1. List some issues of multimedia mining. Describe how back propagation is used in classification. [3 + 7]

2. Describe how bitmap and join indexing are used to represent OLAP data. Explain the different components of data warehouse. [4 + 6]

3. Give any two types of association rules with examples. Trace the results of using the Apriori algorithm on the grocery store example with support threshold 2 and confidence threshold 60%. Show the candidate and frequent itemsets for each database scan. Enumerate all the final frequent itemsets. Also indicate the association rules that are generated. [4 + 6]

Transaction_ID	Items
T1	HotDogs, Buns, Ketchup
T2	HotDogs, Buns
T3	HotDogs, Coke, Chips
T4	Chips, Coke
T5	Chips, Ketchup
T6	HotDogs, Coke, Chips

Group B

Attempt any eight questions. [8×5=40]
Question No. 13 is compulsory.

4. What is the purpose of cluster analysis in data mining? Explain. [5]

5. How does KDD differ with data mining? Describe the stages of data mining. [1 + 4]

6. Explain OLAP operations with examples. [5]

7. Explain the primitives of data mining query language. [5]

8. How different schema are used to model data warehouse? Explain. [5]

9. Describe the significances of pre computation of data cube. [5]

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10. How text mining can be used in social networking? Give your own opinion. [5]

11. Describe any five applications of data mining. [5]

12. How ID3 algorithm can be used as attribute selection while building decision tree? Explain. [5]

13. Write short notes on (**Any TWO**): [2.5 + 2.5]
    a. Outlier Analysis
    b. Data warehouse back end tools
    c. Virtual data warehouse
    d. Market basket analysis

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